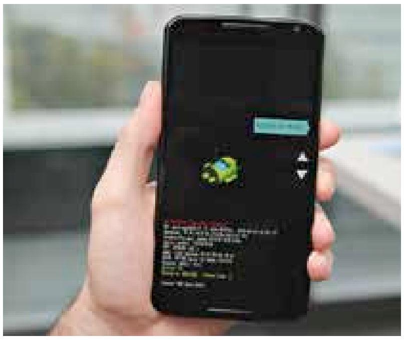
the entire process. Some kernels come pre-installed with an app, so use that if it is available.

HOW TO OVERCLOCK YOUR ANDROID DEVICE

Firstly, you will have to have custom recovery installed, in addition to having a rooted Android device, since you cannot flash kernel utilising stock recovery. Before beginning the process, make sure to have a backup of all the data on your device. You can achieve this either through your custom recovery, apps or various other PC-based solutions. Let's get to business now.

- Download your chosen kernel and all the software that comes along with it. You can also download the kernel to your PC and then import it the zip file to your phone's root directory.
- Next, turn off your device and enter fastboot. This is usually achieved by pressing the power and volume down button at the same time. However, if it doesn't work for your device, a simple Google search should do the trick.
- Use volume keys to navigate in the fastboot screen and move over to Recovery Mode. Select it by clicking the power button. You will enter your custom recovery, for this tutorial, we are using the example of TWRP.
- Click on Install (or Install zip for CWM recovery) and then go over to the location where you stored the kernel.
- Next, you will need to flash the kernel zip file and wait for a success message to pop-up.
- Once the kernel has been flashed successfully, wipe the cache clean.



Recovery Mode on fastboot screen.

- Navigate to Advanced Settings in recovery and click on Fix Permissions.
- Lastly, reboot your system.

Some super user-friendly kernels allow you to set your CPU clock speed during installation itself. You can choose to make changes, if offered, there or later by using the CPU manager app that you have installed. Read on for instructions for the latter.

ALTERING YOUR CPU CLOCK SPEED

If you did not have the ability of changing your device's CPU clock speed during installation, you can now launch your CPU manager app. If you skipped this step, don't worry, simply head over to Play Store and install a reputable app for this purpose now. We recommend Kernel Adiutor, a great app that manages kernel parameters. You will need to grant the GPU manager app/kernel root permission and you may need to install BusyBox as well.

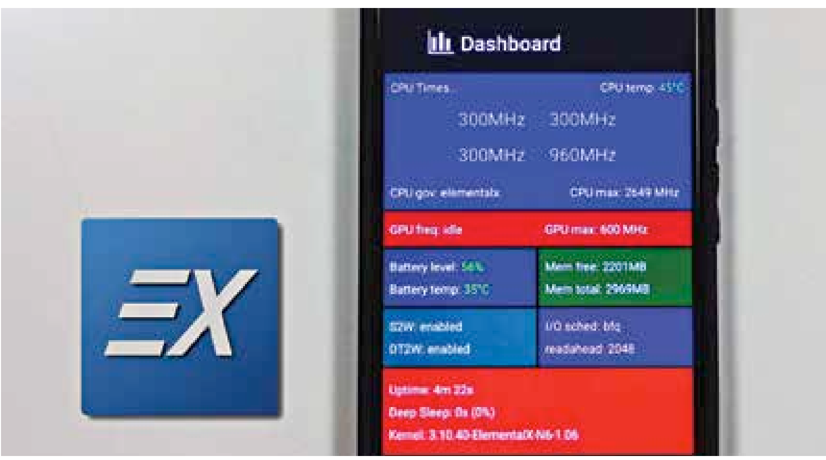
After launching the app, locate the settings that control CPU clock speed. You can now go ahead and choose a max CPU speed from a drop-down menu or slider, depending on the app you are using for this. When we said "max" CPU speed, we didn't mean THE maximum clock speed available. We meant, choose a speed that would be the maximum your CPU would run at. Take this slowly, increase your CPU clock speed by around 10 per cent every time and gauge how it impacts your device. Work your way up only if you need to and there aren't any negative signs such as excessive overheating and battery drainage.

Note that your device will not instantaneously jump in speed with the new settings. Overclocking essentially increase the CPU's maximum speed, to be able to deliver better performance when necessary. If you encounter any kind of instability, like the aforementioned (heating and battery drainage), you can easily revert to the previous CPU clock speed setting (lower) and analyse how this affects the device.

Additionally, if there are any unforeseen issues or predicaments during this process that don't solve themselves by simply dropping the CPU clock speed, you can hit the forums or discussion groups. You can read through the massive amounts of previously-posted questions or simply ask the question yourself. Sites like XDA-developers and Reddit has exceedingly active forums that constantly attempt to answer users and solve any pressing doubts they may have about a large variety of topic.

If nothing is working, you will be better off flashing back your original custom kernel since nothing else has worked in your favour. You can then try with a different custom kernel if you haven't been deterred yet.

You have now successfully managed to overclock your phone. While overclocking comes with quite a few risks, it may very well be worth it. This is because most people would only want to overclock old, dilapidated phones that anyway seem to be dying out. There are a few people that daringly root and overclock brand new phones as well, but we don't recommend that especially since the phones being released today are capable of doing well without any external assistance. However, if you have an old phone that has nearly died out, chances are that overclocking may breathe some life into it.



ElementalX kernel - a well-renowned kernel.